

1: What is a Retain Cycle and How Does One Break It?

Retain Cycle: 2 objects hold strong references to each other. Neither object's reference count can reach 0, so neither is deallocated, causing memory leak(s)

customer——strong——>account

 /\ |
 | ←-----strong-----\ /

If you want to break it, at least one reference must be made weak: ex:

weak var owner: Customer?

It becomes:

customer——strong——>account

 |
 | weak
 |
 Customer

2: The 3 Tries:

try: Calls a throwing function that requires the programmer to handle or propagate it (throw the error)

try?: Converts an error into a nil; error handling, making it nil

try!: Assumes the call succeeds, crashes at runtime if thrown

3: Protocols instead of Base Classes

Protocols are used when defining capabilities that unrelated types can adopt. For example, many items in a program need display for different reasons. Some need display for balance statements, some for account information, and others just to show a meme.

Base Classes are for when you need proper inheritance relationships, a sub item of the big one.

Example: Automobile base class with Lifted Oversized Compensation Impractical Pickup Truck as the subclass

4: Constraints to Add to Generics When Comparing/Sorting Elements

Use comparable constraint: as it means that your item would need to use certain elements that are not always anticipate (compare/sort)